

MANQI LUO

Shanghai Jiaotong University
UC Santa Cruz
lzfsylyyx@gmail.com

EDUCATION

Shanghai Jiao Tong University, China

Sep 2022 – Jun 2026

B.S. in Electronic Engineering
Overall GPA 3.54/4
Average Mark 85.1/100

University of California, Santa Cruz

Expected Sep 2026

Ph.D. in Computer Science and Engineering
Incoming Ph.D. Student

Santa Cruz, USA

Selected Coursework

- C++ Programming: 92
- Data Structures: 97
- Discrete Mathematics: 98
- Machine Learning: 85
- Video Coding and Communication: 90

EXPERIENCE

System Wisdom Lab (SysdomLab), Shanghai Jiao Tong University

Sep 2023 – Jun 2026

Advisor: Prof Chen Chen (chen-chen@sjtu.edu.cn)

- Conduct research on large language models (LLMs) with a focus on optimizing scheduling algorithms for the vLLM system. Adapted the cluster fair queuing algorithm (originally proposed for network research) to the vLLM system, achieving simultaneously both fair and efficient scheduling of LLM tasks.
- Overcame the efficiency bottlenecks of app-level fair sharing methods, such as VTC (virtual token counter) and the fairness limitations of request-level methods, such as FCFS (First Come First Serve). Implemented a priority-based scheduling algorithm driven by relative completion order, allowing high-priority apps to maximize resource usage. Established a relaxed fairness constraint where actual completion times are guaranteed to match or exceed ideal fair distribution outcomes, significantly improving system throughput.
- Addressed the following challenge: How to describe the service costs of LLM applications; How to predict the length of the output token as accurately as possible.
- This paper was accepted by **CloudCom 2025** and was awarded the '**Excellent Paper Award**'.

Remote Internship, University of North Carolina at Chapel Hill

Aug 2025 – Oct 2025

Advisor: Prof Fan Yao (fanyao@unc.edu)

- Optimized the DPO-based preference-alignment training framework.
- Calculated the pairwise loss based on the chosen/rejected pairs of each sample. Using the embeddings of the chosen and rejected responses, a second-order term is added to refine the Bradley-Terry model to better capture the pairwise preference data. Dynamically attenuates the loss for pairs where the chosen and rejected responses exhibit high similarity in their embeddings.

School of Computing Summer Workshop 2023, National University of Singapore

May 2023 - Jul 2023

- Participated in "Structure and Interpretation of Computer Programs" and led my team members to successfully complete the entire project.
- Completed the full development cycle successfully and contributed the overall architectural design and core implementation of the final project.

China Undergraduate Mathematical Contest in Modeling

September 2023

- Won the third prize in Shanghai in **China Undergraduate Mathematical Contest in Modeling**. This project transforms a real engineering project into a mathematical model and then optimizes it.
- Served as the team's modeler, undertaking the abstraction of the real-world problem into a mathematical model while also coordinating and managing a three-member team throughout the competition.

PUBLICATIONS

Yang, M., Wang, G., **Luo, M.**, Liu, Y., Chen, C., Zhao, H., ... Guo, M. (2025). **Justitia: Fair and Efficient Scheduling for LLM Applications**. arXiv preprint arXiv:2510.17015. In Proceedings of The 16th IEEE International Conference on Cloud Computing Technology and Science (CloudCom 2025). (Accepted; **Excellent Paper Award**).

LANGUAGE & TECHNICAL STRENGTHS

Languages:

- Mandarin (native)
- English: TOEFL: 95, GRE: V: 154, Q: 168

Technical Strengths:

- Have extensive experience in optimizing low-level scheduling and reasoning mechanisms of LLMs, as well as in training preference-alignment models using DPO.
- Programming: Python, MATLAB, C++
- Tools: VS Code, Linux, PyTorch